Does Applause Move Stocks?

Audience reaction at Big Tech keynotes and short-run abnormal returns · Taehyeon Kim · September 2026

ABSTRACT

Technology keynotes are staged to create enthusiasm, and their loudest moments are replayed across financial media. This project asks whether that enthusiasm tells investors anything. I catalogue 345 keynotes held by nine large technology companies between 2010 and 2026, measure audience applause second by second from the broadcast audio with a pretrained sound-event detection network (validated against human caption tags; held-out ROC-AUC 0.91), and relate within-company variation in applause to daily abnormal returns. In a pre-registered specification over 250 live-audience keynotes, one standard deviation more applause goes with a change of +0.002 standard deviations in the two-day standardised abnormal return (95% CI -0.15 to +0.15; wild cluster bootstrap p = 0.979). No secondary outcome, alternative applause measure, robustness check or placebo changes this conclusion: applause is not a return signal. Keynotes themselves barely move prices on average, but the stocks rise +1.3% over the ten sessions before them.

345
keynotes catalogued, nine companies, 2010–2026

250
live-audience keynotes in the main sample

0.91
detector ROC-AUC vs human caption tags (held-out videos)

+0.002
SCAR[0,+1] per 1 SD more applause (p = 0.979)

1. Introduction

Every year Apple, Samsung, Google, NVIDIA and their peers spend heavily on keynotes that are built around a handful of ovation moments: the "one more thing", the price reveal, the demo that works. Clips of the loudest reactions circulate within minutes, and commentators read the room as a verdict on the products. For an investor the question is simple: does the room know something the market does not? If audience enthusiasm reflects how compelling the announcements are, louder keynotes should be followed by higher abnormal returns. If it is staged hype, the loudest keynotes may mark expectations that are already priced, and they could even precede disappointment ("buy the rumour, sell the news"). If it is noise—a function of the venue, the audience and the audio mix—there should be no relation at all.

Answering this needs three things that do not exist off the shelf: a complete list of keynotes with precise start times (whether the market was open decides which day reacts), a reliable measure of applause for each keynote, and an event-study design that separates the keynote's effect from the market's. This project builds all three: a hand-curated catalogue of 345 keynotes; an audio pipeline that turns 327 broadcast recordings into second-level applause timelines and is validated against human-written caption tags; and a daily event study with a pre-registered cross-sectional test and inference that respects the small number of companies. It also treats the pandemic years, when keynotes went virtual, as a no-applause comparison group.

2. Data

2.1 The keynote catalogue

The sample is every main keynote held by nine large technology companies between January 2010 and August 2026: Apple (WWDC and product events), Samsung Electronics (Galaxy Unpacked), Alphabet (Google I/O, Made by Google and other launches), Microsoft (Build, Ignite and device events), Meta (F8 and Connect), Amazon (AWS re:Invent CEO keynotes and device events), NVIDIA (GTC, CES and Computex keynotes), Tesla (unveilings, AI days, investor days and shareholder meetings) and AMD (CES, Computex and launch events). Each event was researched by hand: local start time and time zone (which decide whether the market could react the same day), whether an in-person audience was present, and a YouTube video carrying the full broadcast audio, preferring the company's own channel. Manual decisions—dropping out-of-scope events, rejecting videos with commentary, dubbed audio or audience-camcorder sound—are logged in `data/catalog/curation_log.md`. The catalogue holds 345 keynotes, 327 of them with a usable video.

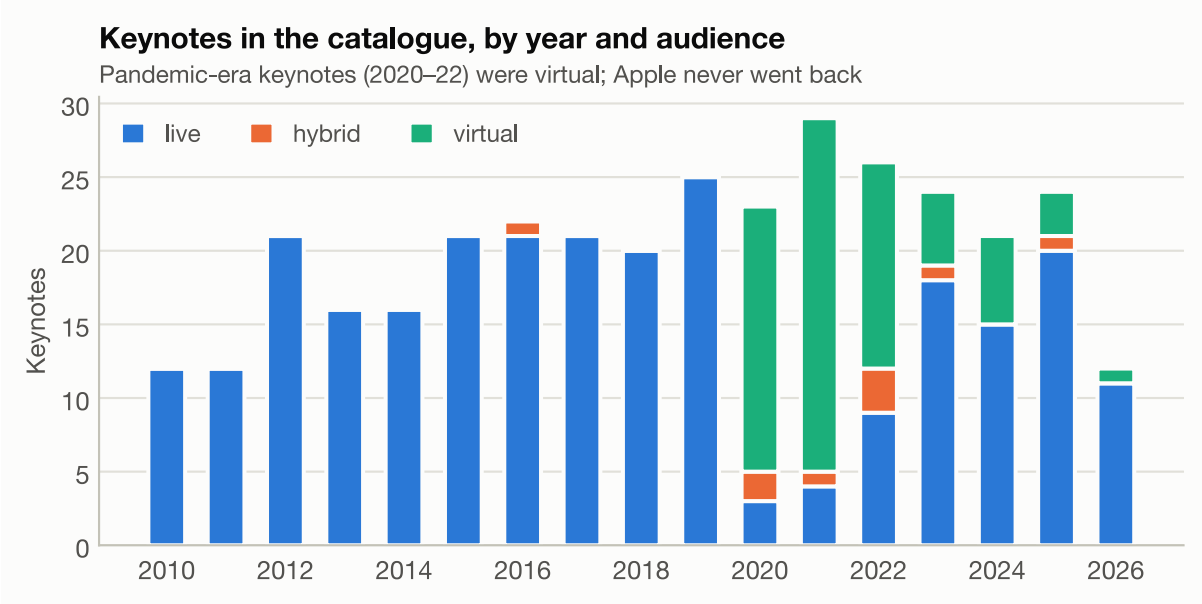


Figure 1. Keynotes per year by audience type. The pandemic moved every company online in 2020–2022; Apple kept pre-recorded keynotes afterwards, which removes its recent events from the live-audience sample.

2.2 From audio to applause

Each video's audio track (the original-language track; machine-dubbed and "stable volume" variants are excluded) is decoded to 32 kHz mono and passed through PANNs `Cnn14_DecisionLevelMax` (Kong et al., 2020), a convolutional network trained on the 527 classes of Google's AudioSet. The network outputs a probability for every class every 0.32 seconds; the pipeline keeps thirteen classes (speech, applause, clapping, cheering, laughter, music and others) averaged to one row per second, stored as a 66 kB file per keynote. The clap score of a second is max(P(applause), P(clapping)).

Two steps turn this timeline into keynote-level measures. First, the *keynote span* keeps the speech-dominated stretch of the video, which removes countdown pre-rolls and post-show music, and it is cut where a closing live-music performance begins (Apple booked musicians for seven of its keynotes). Second, an *ovation* is a run of seconds whose clap score stays above an offset threshold, reaches an onset threshold at least once and lasts at least a minimum duration. The primary measure is **applause share**, the fraction of keynote sec-

onds spent in ovations; secondary measures are ovations per minute, the longest ovation, the mean clap probability (threshold-free) and the share of cheering.

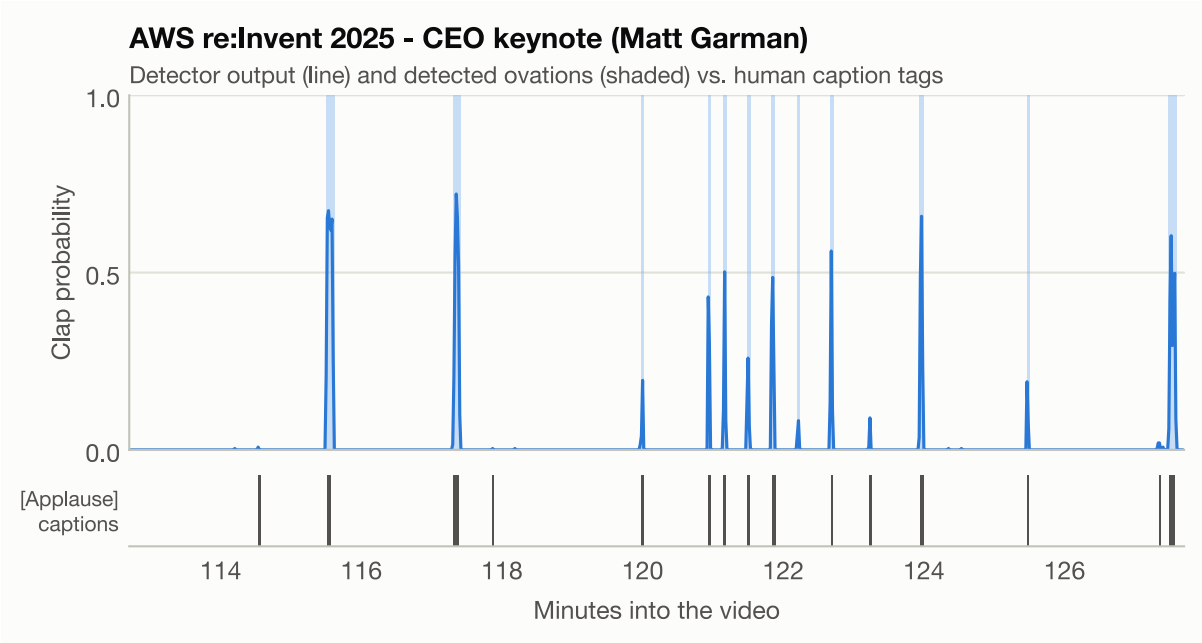


Figure 2. Fifteen minutes of one keynote as the pipeline sees it: the per-second clap score, the detected ovations (shaded) and, underneath, the [Applause] tags a human captioner wrote. The two sources were produced independently. The keynote shown is the caption-labelled video whose detection accuracy (ROC-AUC) is closest to the median, not the best one.

2.3 Validating the detector

Many official uploads carry captions written by professional captioners, who mark audience reactions with tags such as `[Applause]` or `[Cheering and applause]`. These tags are an independent, human label of when the audience clapped. Of the videos with human captions, 22 were used to choose the episode thresholds (onset 0.05, offset 0.02, minimum 2 s) by maximising episode-level F1 with a 2-second tolerance, and 17 were held out. On the held-out videos the detector reaches a second-level ROC-AUC of 0.91, episode precision of 0.76 and recall of 0.92. Caption tags are incomplete—captioners skip short or overlapping applause—so measured precision is a lower bound. As a negative control, keynotes without an in-person audience should show almost no applause, and they do (Figure 3, right).

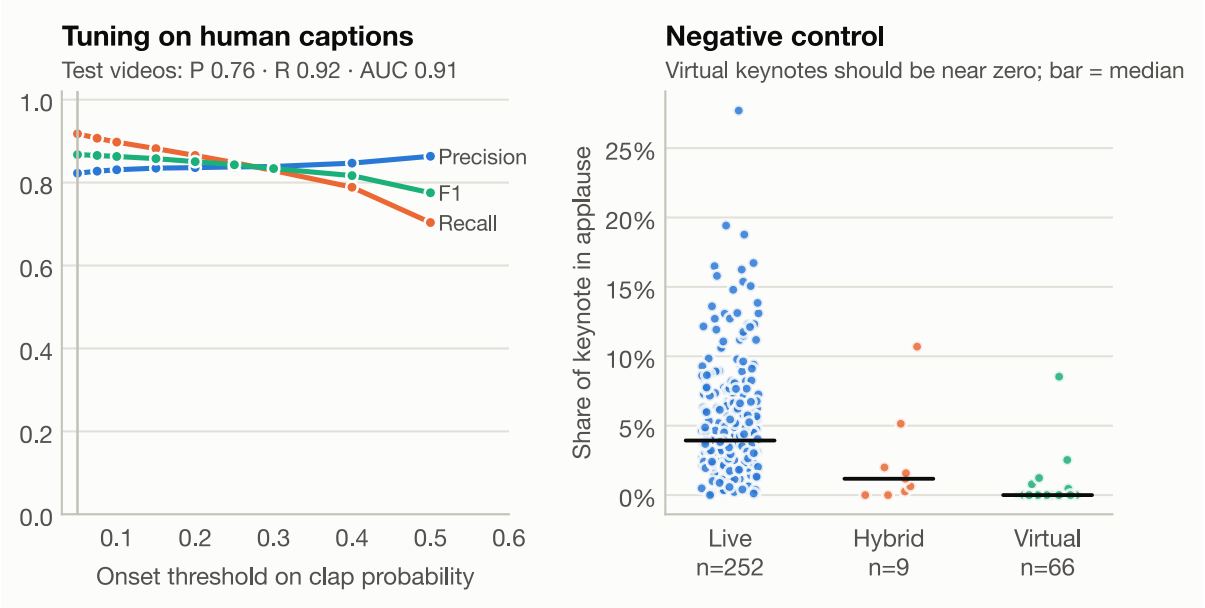


Figure 3. Left: precision, recall and F1 of ovation detection on the development videos as the onset threshold varies (vertical line = chosen threshold). Right: applause share by audience type; virtual keynotes are the negative control.

2.4 Market data

Daily split- and dividend-adjusted prices and volumes come from Yahoo Finance for the nine stocks (Samsung Electronics as 005930.KS on the Korea Exchange), the S&P 500, the Nasdaq-100, KOSPI and the VIX; Fama–French daily factors come from Kenneth French's data library, and earnings-announcement timestamps from Yahoo Finance. Exchange calendars (`exchange_calendars`) give each session's closing time, including early closes and the Korea Exchange's 2016 move from a 15:00 to a 15:30 close.

3. Method

3.1 Event study

The event day t_0 is the first trading session whose close is after the keynote starts. A keynote at 10:00 Pacific therefore has t_0 on the same day, while an evening Tesla event, or a Samsung Unpacked in New York (after midnight in Seoul), has t_0 on the next session. Normal returns come from a market model estimated by OLS over sessions [−250, −30] relative to t_0 (at least 120 returns), using the S&P 500 for US listings and KOSPI for Samsung. Abnormal returns are cumulated over event windows (CAR) and standardised by the estimation-window residual volatility, SCAR = CAR / ($\hat{\sigma}_\varepsilon \sqrt{L}$), which puts a Tesla move and a Microsoft move on the same scale. Average effects are tested with the cross-sectional t-test, the Boehmer, Musumeci and Poulsen (1991) standardised test and Cowan's (1992) generalised sign test.

3.2 Does applause explain the reaction?

The analysis plan (`docs/analysis_plan.md`) was committed to git before any applause measure was joined to returns. Its primary specification regresses SCAR[0,+1] on the within-company z-score of applause share with company fixed effects:

$$\mathrm{SCAR}[0,+1]_i = \alpha_{\mathrm{company}(i)} + \beta \cdot z(\text{applause share})_i + \varepsilon_i$$

Standardising within company compares each keynote with the same company's other keynotes, so company style, audience culture and audio mixing cannot drive the result. With only nine companies, conventional clustered standard errors are unreliable, so inference uses the wild cluster restricted bootstrap (Cameron, Gelbach and Miller, 2008) with Webb's six-point weights and 9,999 draws; heteroskedasticity-robust (HC3) intervals are reported alongside. Secondary outcomes and secondary applause measures form two families, each corrected with the Benjamini–Hochberg procedure.

4. Results

4.1 Keynotes on average

Keynotes move prices less than their reputation suggests. Across all 343 keynotes the mean two-day abnormal return is +0.01% (95% CI -0.39% to +0.43%), and fewer than half are positive. The build-up is where the action is: in the ten sessions before a keynote the stocks earn +1.31% of abnormal return (BMP $p < 0.001$; 59% positive), and after the keynote the cumulative path flattens and drifts down (Figure 4). The pattern is the textbook shape of anticipation followed by a muted announcement.

Company averages differ. Apple's keynotes are followed by a mean two-day abnormal return of -0.74% ($p = 0.018$; only 36% positive), and Tesla's by -2.16% ($p = 0.016$), in line with the "sell-the-news" folklore around both companies. These per-company averages are descriptive: nine companies are tested, and Tesla's worst reactions coincide with other news such as delivery reports released over the same weekend.

Table 1. Average abnormal returns around keynotes (market model).

Sample	Window	N	Mean CAR	95% CI	p (t)	p (BMP)	Positive
All keynotes	[0,+1]	343	+0.01%	[-0.39, +0.43]	0.944	0.750	48%
All keynotes	[+2,+20]	340	-0.57%	[-1.54, +0.40]	0.244	0.146	43%
Audience: live	[0,+1]	263	+0.04%	[-0.42, +0.51]	0.873	0.527	48%
Audience: live	[+2,+20]	261	-0.08%	[-1.23, +1.06]	0.888	0.421	44%
Audience: hybrid	[0,+1]	9	+0.11%	[-3.55, +3.88]	0.957	0.833	67%
Audience: hybrid	[+2,+20]	9	-2.00%	[-9.40, +4.99]	0.622	0.881	56%
Audience: virtual	[0,+1]	71	-0.09%	[-0.82, +0.67]	0.822	0.758	48%
Audience: virtual	[+2,+20]	70	-2.22%	[-3.84, -0.72]	0.008	0.064	37%

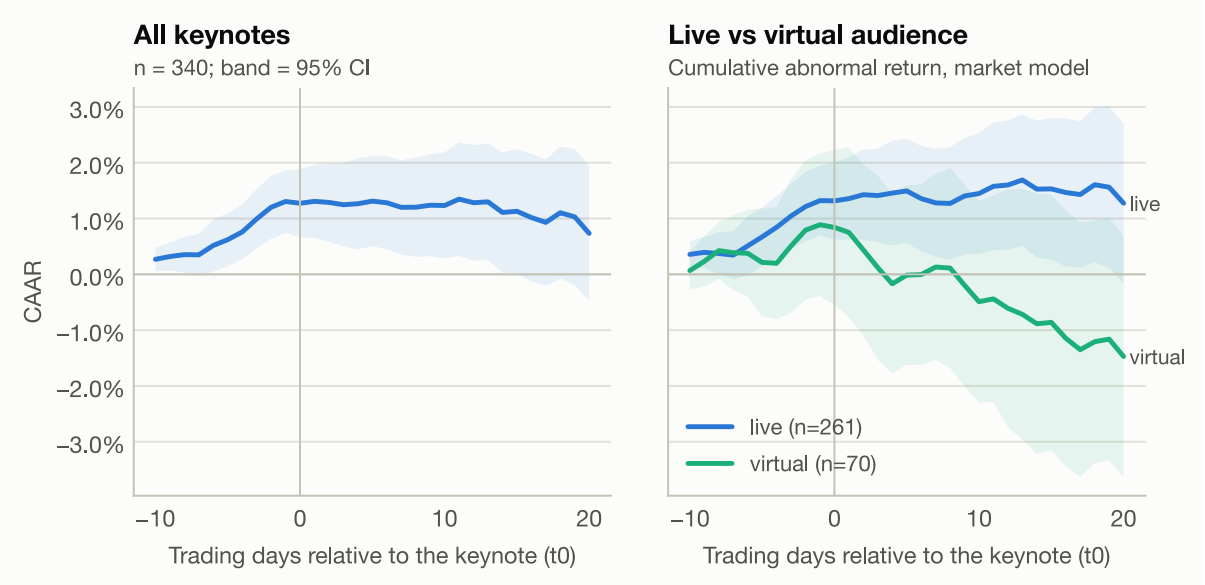


Figure 4. Cumulative average abnormal return from ten sessions before to twenty sessions after the keynote, with 95% confidence bands.

4.2 Applause and the stock's reaction

The pre-registered test finds no relation between applause and the stock's reaction. The coefficient on the within-company z-score of applause share is +0.002 (HC3 95% CI -0.148 to +0.152; wild cluster bootstrap p = 0.979; N = 250 keynotes in 9 companies). The within-company rank correlation is -0.03 (permutation p = 0.670).

The estimate is not just insignificant; it is small and precise. Because the outcome is standardised, the interval says that one standard deviation more applause shifts the two-day abnormal return by at most about 0.15 residual standard deviations—roughly 0.3 percentage points for a typical stock in the sample. The secondary families agree (Table 2): neither the direction of the move, its size, the post-event drift nor abnormal trading volume is related to applause, and neither are the alternative measures (the smallest Benjamini–Hochberg q-value is 0.84).

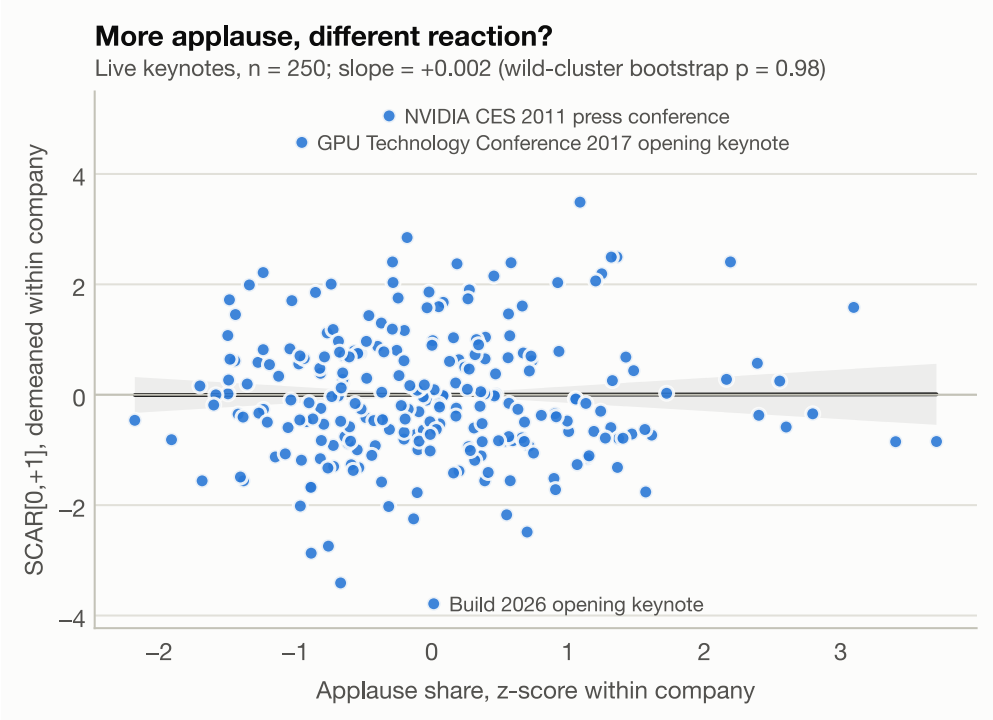


Figure 5. Within-company applause share against SCAR[0,+1] (both demeaned within company, so the line's slope is the fixed-effects estimate). Shaded band: 95% HC3 interval of the slope.

Table 2. Pre-registered secondary analyses (company fixed effects; q = Benjamini–Hochberg adjusted within each family).

Family	Outcome / measure	β	95% CI (HC3)	p (WCB)	q	N
Primary	**SCAR[0,+1] ~ applause share**	**+0.002**	**[-0.148, +0.152]**	**0.979**	**–**	**250**
Outcome	SCAR[0,0]	+0.066	[-0.096, +0.228]	0.516	0.84	250
Outcome	SCAR[-1,+1]	+0.011	[-0.127, +0.149]	0.909	0.91	250
Outcome	SCAR[0,+5]	+0.050	[-0.116, +0.217]	0.451	0.84	250
Outcome	SCAR[+2,+20]	+0.018	[-0.136, +0.171]	0.703	0.84	248
Outcome	|SCAR[0,+1]|	-0.025	[-0.120, +0.070]	0.542	0.84	250
Outcome	Abnormal log volume [0,+1] (std.)	+0.050	[-0.082, +0.181]	0.606	0.84	250
Measure	Ovations per minute	-0.006	[-0.155, +0.144]	0.939	0.95	250
Measure	Longest ovation (log seconds)	+0.041	[-0.113, +0.194]	0.733	0.95	250
Measure	Mean clap probability	-0.006	[-0.159, +0.147]	0.953	0.95	250
Measure	Cheering share	-0.076	[-0.212, +0.061]	0.256	0.95	250
Measure	Hype index (1st principal component)	-0.007	[-0.153, +0.138]	0.944	0.95	250

SCAR outcomes are in units of estimation-window residual standard deviations per one within-company standard deviation of applause share. WCB = wild cluster restricted bootstrap over companies.

4.3 Sorting, robustness and placebo

Sorting keynotes into within-company terciles of applause shows no monotone pattern: the loudest third minus the quietest third is -0.05 standard deviations (95% CI -0.43 to +0.34; Figure 6). Across every alternative

normal-return model, control set and subsample—including the exploratory exclusions of keynotes near an earnings release and of keynotes with uncertain start times—the coefficient stays between -0.05 and +0.08, with no bootstrap p-value below 0.38 (Figure 7).

The placebo test pairs each keynote's applause with the stock's reaction on a random non-event day of the same company. The actual coefficient sits in the middle of the placebo distribution (97% of placebo draws are at least as extreme; Figure 8), which is what one would expect if applause carried no information about keynote-day returns. Company by company (Figure 9) the slopes scatter around zero with no company showing a clear relation.

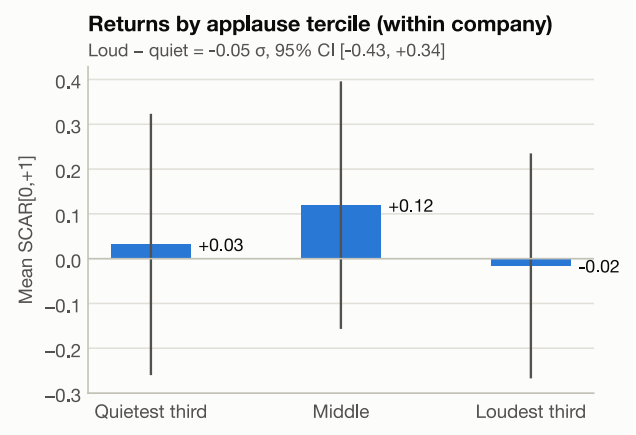


Figure 6. Mean SCAR[0,+1] by within-company applause tercile, with 95% intervals.

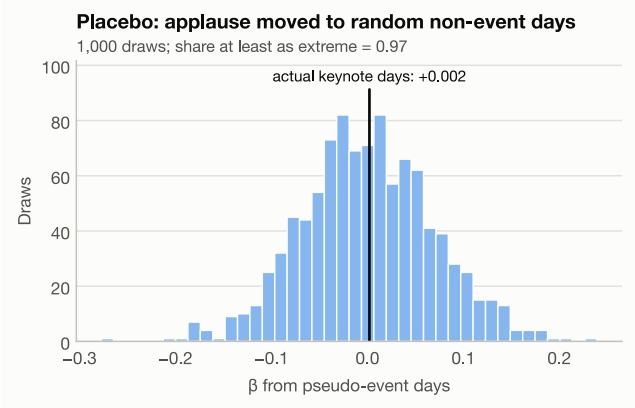


Figure 8. The coefficient when each keynote's applause is paired with the stock's reaction on a random non-event day of the same company.

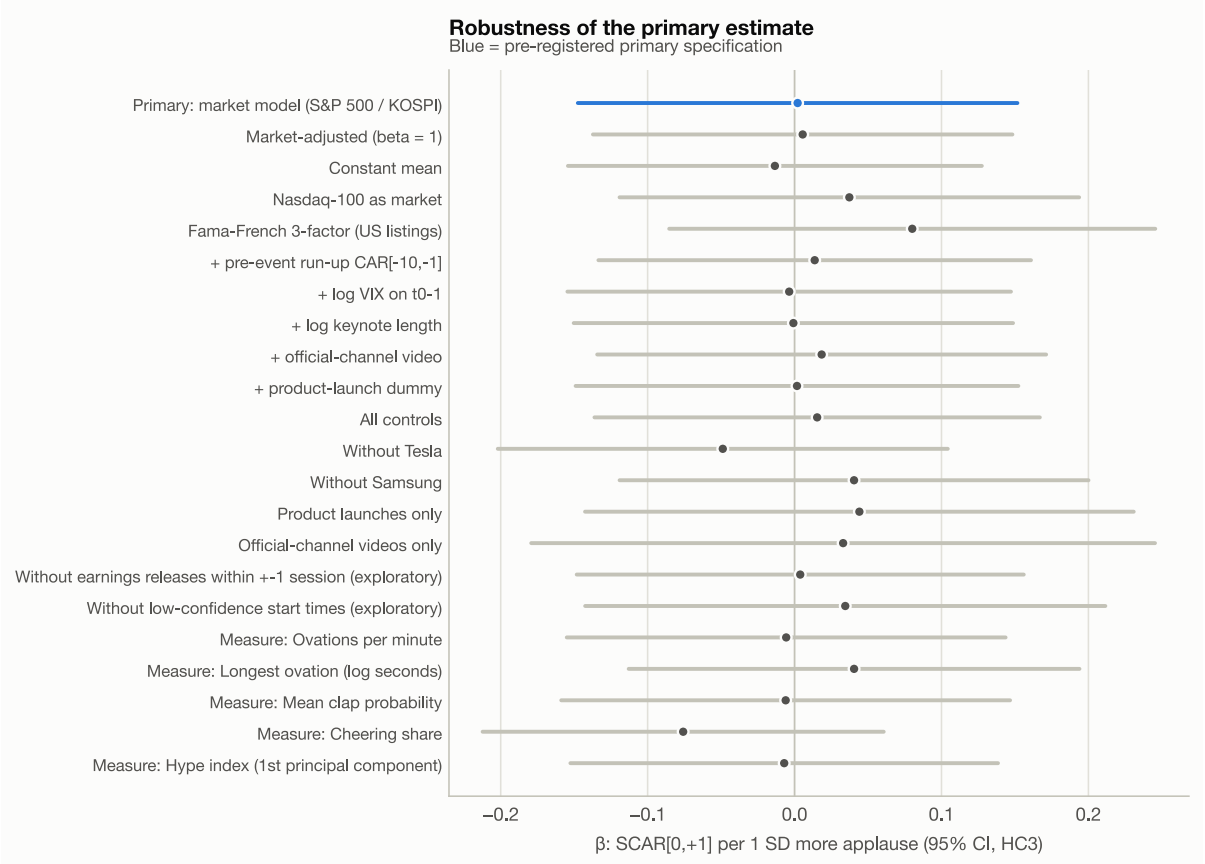


Figure 7. The applause coefficient across normal-return models, controls, subsamples and alternative applause measures (95% HC3 intervals). Rows marked exploratory were added after the plan was registered.

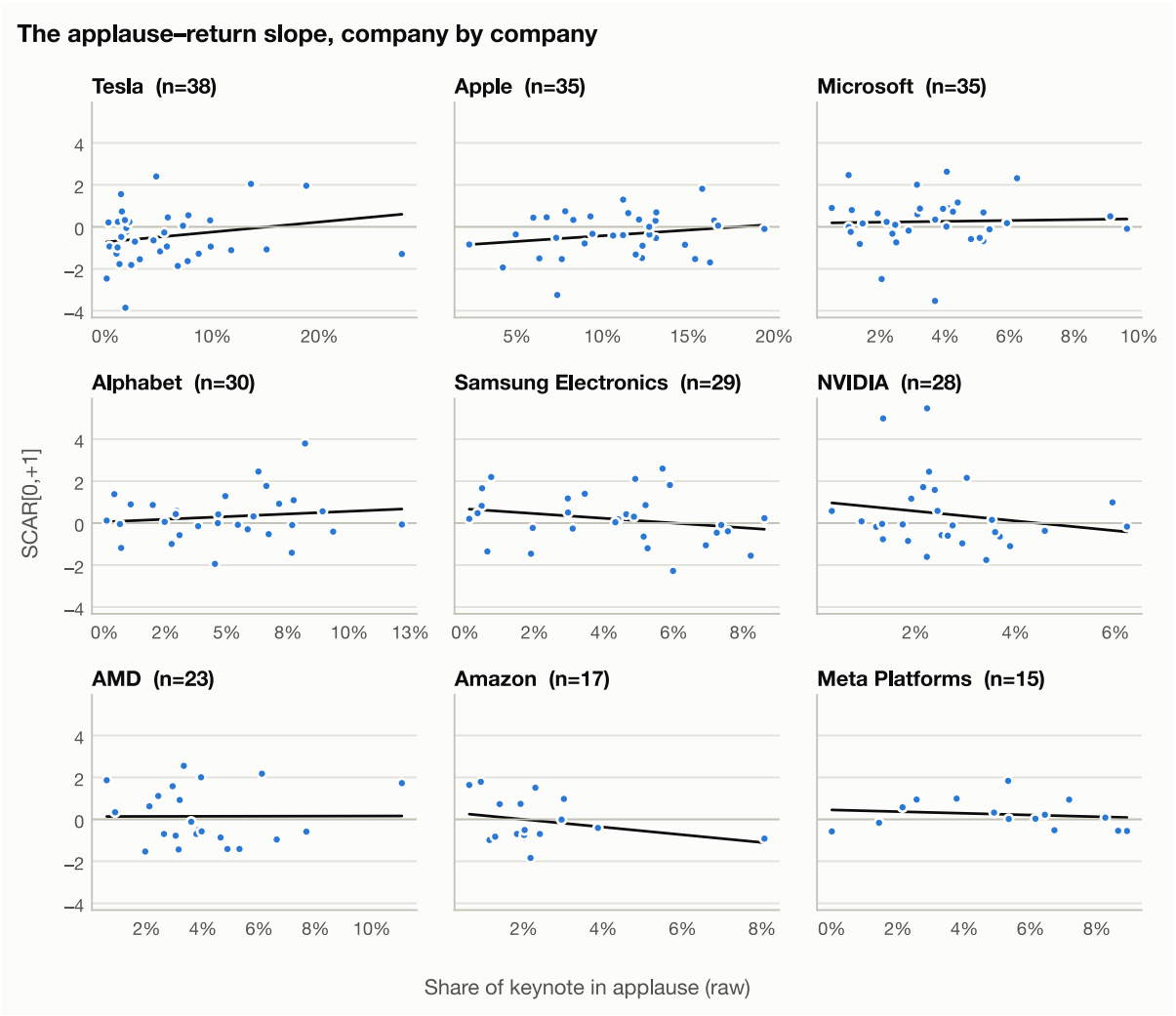


Figure 9. Raw applause share against SCAR[0,+1], company by company.

4.4 Live versus virtual keynotes

Virtual keynotes offer a crude natural experiment: no audience, no applause. They are followed by reactions that are statistically indistinguishable from live keynotes in direction, size and trading volume (Table 3). Because virtual keynotes cluster in 2020–2022, this comparison also carries the pandemic market, so it is descriptive only.

Table 3. Live-audience versus virtual keynotes (descriptive).

Outcome	Live	Virtual	Difference	p (Welch)	N live / virtual
SCAR[0,+1]	+0.049	-0.052	+0.102	0.586	263 / 71
|SCAR[0,+1]|	+0.948	+1.079	-0.131	0.282	263 / 71
Abnormal log volume [0,+1] (std.)	+0.194	+0.306	-0.111	0.482	263 / 71

5. Discussion and limitations

Why is the applause signal empty? Three explanations fit the evidence. First, keynote audiences are not a random sample of investors: they are employees, developers, partners and fans, and companies script the moments that invite applause, so the volume of the room reflects the production as much as the products. Second, the information that matters— specifications, prices, launch dates, AI capabilities—is public the moment it is spoken and is analysed by professionals within minutes; the room's reaction adds little beyond it. Third, much of the news is anticipated: the significant pre-keynote run-up suggests that expectations are formed, and priced, before anyone claps.

The null is informative because the design had power. With 250 live keynotes, the confidence interval excludes effects that would matter for trading, and the detector's accuracy against human labels makes measurement error an unlikely explanation.

The study has limits. Daily returns blur what happens during the keynote itself; minute-level prices around keynotes held during market hours would sharpen the test. Applause is measured on published videos whose audio mix varies across productions; within-company standardisation and the official-channel robustness check address this but cannot remove it. Caption tags are an imperfect ground truth (one Apple caption track does not line up with its video at all). Some start times are uncertain, some event windows overlap for the same company, and other news occasionally lands inside the window. Finally, the sample is nine of the most liquid stocks in the world, where short-lived signals are least likely to survive; smaller companies' launch events might behave differently.

6. Conclusion

Audience applause at tech keynotes is easy to measure accurately from audio, but it does not predict how the company's stock reacts: louder keynotes are followed neither by higher nor by lower abnormal returns, nor by larger moves or heavier trading. The market's reaction to keynotes happens mostly before them. For practitioners the result is a useful negative: the loudest clip of a keynote is marketing, not a signal.

References

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Appendix A. Live-audience keynotes in the main sample

Date	Company	Keynote	Applause share	z	CAR[0,+1]	SCAR[0,+1]
2010-01-27	Apple	Apple Special Event (iPad introduction)	7.6%	-0.82	-3.08%	-1.54
2010-02-14	Samsung Electronics	Samsung Mobile Unpacked 2010 (MWC Barcelona)	0.5%	-1.44	+1.64%	+0.82
2010-04-08	Apple	Apple Special Event (iPhone OS 4 preview)	4.9%	-1.49	-0.69%	-0.36
2010-05-19	Alphabet	Google I/O 2010 Day 1 Keynote	2.7%	-0.72	-1.41%	-0.99
2010-06-07	Apple	WWDC 2010 Keynote	12.3%	+0.36	-2.71%	-1.49
2010-09-01	Apple	Apple Special Event (fall music event: iPod	9.4%	-0.37	-0.64%	-0.33
2010-09-08	Alphabet	Google Instant launch event (search event)	1.1%	-1.27	+1.50%	+0.89
2010-10-20	Apple	Back to the Mac	2.2%	-2.18	-1.52%	-0.84
2010-12-07	Alphabet	Google Chrome event (Chrome Web Store & Chro	2.0%	-0.97	+1.69%	+0.86
2011-01-05	NVIDIA	NVIDIA CES 2011 press conference	2.2%	-0.31	+17.47%	+5.47
2011-02-02	Alphabet	Android 3.0 Honeycomb event	0.6%	-1.42	-0.09%	-0.05
2011-02-13	Samsung Electronics	Samsung Mobile Unpacked 2011 (MWC Barcelona)	0.5%	-1.44	+2.71%	+1.66
2011-03-02	Apple	Apple Special Event (iPad 2)	6.0%	-1.23	+0.73%	+0.44
2011-05-10	Alphabet	Google I/O 2011 Day 1 Keynote	12.1%	+2.40	-0.11%	-0.07
2011-06-06	Apple	WWDC 2011 Keynote	15.4%	+1.14	-2.33%	-1.53
2011-09-01	Samsung Electronics	Samsung Mobile Unpacked 2011 Episode II (IFA	0.1%	-1.58	+0.41%	+0.20
2011-09-13	Microsoft	BUILD 2011 day 1 keynote	5.9%	+1.05	+0.23%	+0.18
2011-10-04	Apple	Let's talk iPhone	4.2%	-1.68	-2.89%	-1.94
2011-10-19	Samsung Electronics	Samsung Mobile Unpacked 2011 Google Episode	0.8%	-1.34	+4.47%	+2.20
2012-01-09	NVIDIA	NVIDIA CES 2012 press conference	2.9%	+0.20	-3.77%	-0.97
2012-01-19	Apple	Apple Education Event (iBooks textbooks)	6.3%	-1.15	-2.66%	-1.51
2012-02-09	Tesla	Model X Unveil	9.8%	+0.68	-3.60%	-0.94
2012-02-29	Microsoft	Windows 8 Consumer Preview event	0.5%	-1.48	+1.11%	+0.90
2012-03-07	Apple	Apple Special Event (the new iPad)	6.7%	-1.04	+0.79%	+0.45
2012-05-03	Samsung Electronics	Samsung Mobile Unpacked 2012 (London)	0.6%	-1.38	-2.87%	-1.35
2012-06-06	Tesla	2012 Annual Meeting of Stockholders	0.3%	-0.93	+0.85%	+0.21
2012-06-11	Apple	WWDC 2012 Keynote	13.1%	+0.57	-1.13%	-0.53
2012-06-18	Microsoft	Microsoft Surface reveal	3.1%	-0.25	+2.79%	+2.01
2012-06-20	Microsoft	Windows Phone Summit (Windows Phone 8 reveal	1.0%	-1.23	-0.01%	-0.01
2012-06-22	Tesla	Model S Customer Delivery Event	27.7%	+3.70	-5.38%	-1.30
2012-06-27	Alphabet	Google I/O 2012 Day 1 Keynote	9.3%	+1.47	-0.84%	-0.41
2012-08-29	Samsung Electronics	Samsung Mobile Unpacked 2012 (IFA Berlin)	0.4%	-1.49	+1.00%	+0.47
2012-09-06	Amazon	Amazon Kindle press event (Kindle Fire HD an	3.0%	+0.40	+3.08%	+0.97
2012-09-12	Apple	Apple Special Event (iPhone 5)	7.8%	-0.77	+1.61%	+0.74
2012-09-24	Tesla	Supercharger Unveil	2.7%	-0.53	-8.06%	-1.77
2012-10-23	Apple	Apple Special Event (iPad mini)	8.9%	-0.48	-1.60%	-0.79
2012-10-25	Microsoft	Windows 8 launch event	1.9%	-0.81	+0.86%	+0.64
2012-10-30	Microsoft	Build 2012 day 1 keynote	4.0%	+0.19	+3.48%	+2.63
2012-11-28	Amazon	AWS re:Invent 2012 - Day 1 keynote (Andy Jas	1.4%	-0.57	+1.91%	+0.72
2013-01-06	NVIDIA	NVIDIA CES 2013 press conference	2.2%	-0.31	-3.93%	-1.61
2013-03-14	Samsung Electronics	Samsung Mobile Unpacked 2013 Episode 1 (New	1.9%	-0.89	-2.48%	-1.46
2013-03-19	NVIDIA	GPU Technology Conference 2013 opening keyno	0.9%	-1.27	+0.20%	+0.08
2013-05-15	Alphabet	Google I/O 2013 Keynote	7.7%	+0.93	+1.77%	+1.09
2013-06-04	Tesla	2013 Annual Meeting of Stockholders	1.5%	-0.72	+2.96%	+0.73
2013-06-10	Apple	WWDC 2013 Keynote	16.5%	+1.42	+0.75%	+0.30
2013-06-20	Tesla	Battery Swap Demonstration	9.8%	+0.67	+1.47%	+0.31
2013-06-26	Microsoft	Build 2013 day 1 keynote	4.1%	+0.20	+1.25%	+0.89
2013-07-24	Alphabet	Breakfast with Sundar Pichai (Nexus 7 & Chro	0.7%	-1.40	-1.83%	-1.19
2013-09-04	Samsung Electronics	Samsung Unpacked 2013 Episode 2 (IFA Berlin)	4.4%	+0.08	+0.26%	+0.18
2013-09-10	Apple	Apple Special Event (iPhone 5s and 5c)	7.3%	-0.88	-8.42%	-3.25
2013-09-23	Microsoft	Surface 2 launch event	2.9%	-0.37	-0.34%	-0.18
2013-10-22	Apple	Apple Special Event (iPad Air)	8.3%	-0.65	+0.87%	+0.33
2013-11-13	Amazon	AWS re:Invent 2013 - Day 1 keynote (Andy Jas	0.9%	-0.85	+3.43%	+1.79
2014-01-05	NVIDIA	NVIDIA CES 2014 press conference	2.4%	-0.20	+2.62%	+1.58
2014-02-24	Samsung Electronics	Samsung Unpacked 5 (Unpacked 2014 Episode 1,	7.6%	+1.31	-0.53%	-0.39
2014-04-02	Microsoft	Build 2014 day 1 keynote	4.8%	+0.54	-1.31%	-0.58
2014-04-30	Meta Platforms	F8 2014	2.6%	-0.96	+4.00%	+0.94

2014-05-20	Microsoft	Surface Pro 3 launch event	3.1%	-0.24	+1.29%	+0.60
2014-06-02	Apple	WWDC 2014 Keynote	13.1%	+0.56	+0.59%	+0.29
2014-06-03	Tesla	2014 Annual Meeting of Stockholders	2.0%	-0.65	-1.23%	-0.23
2014-06-18	Amazon	Amazon Fire Phone launch event	3.9%	+0.91	-0.91%	-0.41
2014-06-25	Alphabet	Google I/O 2014 Keynote	7.1%	+0.74	+1.65%	+0.92
2014-09-03	Samsung Electronics	Samsung Unpacked 2014 Episode 2 (IFA Berlin)	5.2%	+0.39	+1.19%	+0.86
2014-09-09	Apple	Apple Special Event (iPhone 6 and Apple Watc	11.2%	+0.09	+2.58%	+1.30
2014-09-20	Meta Platforms	Oculus Connect (OC1)	3.8%	-0.52	+3.05%	+0.99
2014-10-09	Tesla	Tesla D Event (Dual Motor and Autopilot)	1.3%	-0.76	-8.50%	-1.77
2014-10-16	Apple	Apple Special Event (iPad Air 2)	10.7%	-0.04	-0.87%	-0.47
2014-11-12	Amazon	AWS re:Invent 2014 - Day 1 keynote (Andy Jas	1.9%	-0.26	+1.68%	+0.74
2015-01-04	NVIDIA	NVIDIA CES 2015 press conference	1.3%	-0.95	-1.48%	-0.77
2015-01-21	Microsoft	Windows 10: The Next Chapter	1.1%	-1.20	-0.33%	-0.24
2015-03-01	Samsung Electronics	Samsung Galaxy Unpacked 2015 Episode 1 (MWC	5.9%	+0.67	+3.29%	+1.81
2015-03-09	Apple	Spring Forward	11.2%	+0.08	-0.63%	-0.40
2015-03-17	NVIDIA	GPU Technology Conference 2015 opening keyno	1.8%	-0.59	-1.58%	-0.85
2015-03-25	Meta Platforms	F8 2015 (Day 1)	1.4%	-1.38	-0.37%	-0.17
2015-04-29	Microsoft	Build 2015 day 1 keynote	3.7%	+0.02	+0.51%	+0.35
2015-04-30	Tesla	Tesla Energy Unveil (Powerwall)	7.2%	+0.24	+0.17%	+0.05
2015-05-04	Microsoft	Microsoft Ignite 2015 keynote	1.4%	-1.07	-1.20%	-0.82
2015-05-28	Alphabet	Google I/O 2015 Keynote	6.7%	+0.60	-0.76%	-0.53
2015-06-03	AMD	AMD Press Conference at Computex 2015	2.1%	-0.82	+2.47%	+0.62
2015-06-08	Apple	WWDC 2015 Keynote	12.7%	+0.47	-0.59%	-0.37
2015-06-09	Tesla	2015 Annual Meeting of Stockholders	5.1%	-0.12	-3.73%	-1.17
2015-06-16	AMD	AMD E3 2015 'A New Era of PC Gaming' Live Ev	2.9%	-0.46	+6.23%	+1.58
2015-08-13	Samsung Electronics	Samsung Galaxy Unpacked 2015 Episode 2 (New	3.0%	-0.48	+1.02%	+0.50
2015-09-09	Apple	Hey Siri, give us a hint	11.5%	+0.16	+1.05%	+0.65
2015-09-24	Meta Platforms	Oculus Connect 2 (OC2)	8.9%	+1.37	-1.01%	-0.56
2015-09-29	Alphabet	Google Nexus event (September 2015 press eve	0.1%	-1.60	+0.26%	+0.12
2015-09-29	Tesla	Model X Launch Event	2.4%	-0.57	-5.60%	-1.82
2015-10-06	Microsoft	Windows 10 Devices event	9.6%	+2.79	-0.16%	-0.09
2015-10-07	Amazon	AWS re:Invent 2015 - keynote (Andy Jassy)	1.1%	-0.72	-2.77%	-0.99
2016-02-21	Samsung Electronics	Samsung Galaxy Unpacked 2016 (MWC Barcelona)	3.1%	-0.42	-0.57%	-0.26
2016-03-21	Apple	Let us loop you in	9.3%	-0.39	+0.86%	+0.50
2016-03-30	Microsoft	Build 2016 day 1 keynote	5.9%	+1.06	+0.31%	+0.18
2016-03-31	Tesla	Model 3 Unveil	13.8%	+1.36	+6.95%	+2.05
2016-04-05	NVIDIA	GPU Technology Conference 2016 opening keyno	1.2%	-1.05	-0.50%	-0.18
2016-04-12	Meta Platforms	F8 2016 (Day 1)	0.0%	-1.91	-1.34%	-0.58
2016-05-06	NVIDIA	GeForce GTX 1080 launch event	6.2%	+2.60	-0.45%	-0.17
2016-05-18	Alphabet	Google I/O 2016 Keynote	5.8%	+0.31	-0.65%	-0.30
2016-05-31	Tesla	2016 Annual Meeting of Stockholders	2.8%	-0.52	-2.49%	-0.71
2016-06-01	AMD	AMD Press Conference at Computex 2016	1.9%	-0.88	-8.24%	-1.53
2016-06-13	Apple	WWDC 2016 Keynote	12.7%	+0.47	-0.00%	-0.00
2016-08-02	Samsung Electronics	Samsung Galaxy Note Unpacked 2016 (New York)	5.1%	+0.37	-1.27%	-0.64
2016-09-07	Apple	See you on the 7th	12.3%	+0.37	-1.57%	-0.90
2016-09-26	Microsoft	Microsoft Ignite 2016 innovation keynote	2.2%	-0.68	+0.39%	+0.23
2016-10-04	Alphabet	Made by Google 2016 (Pixel launch event)	2.5%	-0.81	+0.09%	+0.06
2016-10-06	Meta Platforms	Oculus Connect 3 (OC3)	6.4%	+0.46	+0.50%	+0.22
2016-10-26	Microsoft	Microsoft Windows 10 event (Surface Studio)	5.2%	+0.72	-0.95%	-0.68
2016-10-27	Apple	hello again	7.3%	-0.90	-0.95%	-0.52
2016-10-28	Tesla	Solar Roof and Powerwall 2 Unveil	11.8%	+1.01	-3.80%	-1.11
2016-11-30	Amazon	AWS re:Invent 2016 - keynote (Andy Jassy)	2.0%	-0.20	-2.01%	-0.87
2017-01-04	NVIDIA	CES 2017 opening keynote (NVIDIA)	3.7%	+0.75	-2.50%	-0.64
2017-03-29	Samsung Electronics	Samsung Galaxy Unpacked 2017 (New York)	7.2%	+1.19	-0.90%	-0.46
2017-04-18	Meta Platforms	F8 2017 (Day 1)	2.1%	-1.12	+0.91%	+0.57
2017-05-02	Microsoft	#MicrosoftEDU event	5.1%	+0.66	-0.59%	-0.53
2017-05-10	Microsoft	Build 2017 day 1 keynote	2.5%	-0.55	-0.83%	-0.74
2017-05-10	NVIDIA	GPU Technology Conference 2017 opening keyno	1.3%	-0.95	+21.29%	+4.99
2017-05-17	Alphabet	Google I/O 2017 Keynote	4.6%	-0.09	-0.00%	-0.00
2017-05-23	Microsoft	Surface Pro launch event Shanghai	5.4%	+0.80	-0.13%	-0.12
2017-05-31	AMD	AMD Press Conference at Computex 2017	3.0%	-0.41	-4.09%	-0.78
2017-06-05	Apple	WWDC 2017 Keynote	10.6%	-0.06	-0.59%	-0.42

2017-06-06	Tesla	2017 Annual Meeting of Stockholders	1.5%	-0.74	+4.44%	+1.56
2017-07-28	Tesla	Model 3 Handover Event	6.8%	+0.16	-5.21%	-1.86
2017-08-23	Samsung Electronics	Samsung Galaxy Unpacked 2017 (Note8, New Yor	6.9%	+1.07	-1.98%	-1.06
2017-09-12	Apple	Apple Special Event (iPhone X)	11.9%	+0.27	-1.75%	-1.32
2017-10-04	Alphabet	Made by Google 2017 (Pixel 2 event)	4.6%	-0.09	+0.49%	+0.42
2017-10-11	Meta Platforms	Oculus Connect 4 (OC4)	4.9%	-0.10	+0.45%	+0.32
2017-11-16	Tesla	Tesla Semi and Roadster Unveil	1.5%	-0.73	-1.45%	-0.48
2017-11-29	Amazon	AWS re:Invent 2017 - keynote (Andy Jassy)	2.2%	-0.11	-2.39%	-1.84
2018-01-07	NVIDIA	NVIDIA CES 2018 press event	2.4%	-0.16	+1.96%	+0.58
2018-02-25	Samsung Electronics	Samsung Galaxy Unpacked 2018 (MWC Barcelona)	4.3%	+0.05	+0.06%	+0.04
2018-03-27	Apple	Let's take a field trip (education event)	14.8%	+1.00	-1.20%	-0.85
2018-03-27	NVIDIA	GPU Technology Conference 2018 opening keyno	3.4%	+0.55	-5.79%	-1.76
2018-05-01	Meta Platforms	F8 2018 (Day 1)	5.3%	+0.05	+2.99%	+1.83
2018-05-07	Microsoft	Build 2018 day 1 keynote	2.4%	-0.56	+0.11%	+0.10
2018-05-08	Alphabet	Google I/O 2018 Keynote	4.9%	+0.01	+1.54%	+1.28
2018-06-04	NVIDIA	NVIDIA COMPUTEX 2018 press conference	0.3%	-1.70	+1.59%	+0.58
2018-06-04	Apple	WWDC 2018 Keynote	13.1%	+0.57	+1.00%	+0.69
2018-06-05	Tesla	2018 Annual Meeting of Stockholders	4.7%	-0.18	+7.84%	+2.40
2018-06-06	AMD	AMD Press Conference at Computex 2018	3.6%	-0.17	-0.45%	-0.12
2018-08-09	Samsung Electronics	Samsung Galaxy Unpacked August 2018 (Brookly	7.4%	+1.24	-0.17%	-0.09
2018-08-20	NVIDIA	GeForce RTX 20 series launch (GeForce Gaming	5.9%	+2.39	+2.39%	+0.99
2018-09-12	Apple	Gather round	12.1%	+0.32	+0.50%	+0.35
2018-09-26	Meta Platforms	Oculus Connect 5 (OC5)	7.2%	+0.73	+2.48%	+0.94
2018-10-09	Alphabet	Made by Google 2018 (Pixel 3 event)	3.1%	-0.61	-0.75%	-0.57
2018-10-30	Apple	There's more in the making	16.7%	+1.48	+0.09%	+0.06
2018-11-06	AMD	AMD Next Horizon	2.4%	-0.68	+4.62%	+1.11
2018-11-28	Amazon	AWS re:Invent 2018 - keynote (Andy Jassy)	0.6%	-1.02	+2.86%	+1.64
2019-01-06	NVIDIA	NVIDIA CES 2019 press event	2.7%	+0.06	-0.37%	-0.11
2019-01-09	AMD	CES 2019 Keynote: Dr. Lisa Su	5.3%	+0.58	-6.98%	-1.41
2019-02-20	Samsung Electronics	Samsung Galaxy Unpacked 2019 (San Francisco)	8.6%	+1.72	+0.41%	+0.23
2019-02-24	Microsoft	HoloLens 2 reveal (MWC 2019)	1.1%	-1.19	+1.00%	+0.80
2019-03-14	Tesla	Model Y Unveil	7.7%	+0.32	-8.49%	-1.63
2019-03-18	NVIDIA	GPU Technology Conference 2019 opening keyno	1.9%	-0.54	+4.23%	+1.16
2019-03-19	Alphabet	Stadia reveal: Google GDC 2019 keynote	6.3%	+0.46	+3.54%	+2.46
2019-03-25	Apple	It's show time	16.2%	+1.36	-3.17%	-1.69
2019-04-22	Tesla	Autonomy Investor Day	0.4%	-0.92	-4.77%	-0.93
2019-04-30	Meta Platforms	F8 2019 (Day 1)	5.3%	+0.06	+0.05%	+0.02
2019-05-06	Microsoft	Build 2019 vision keynote	1.4%	-1.03	+0.20%	+0.16
2019-05-07	Alphabet	Google I/O 2019 Keynote	6.1%	+0.39	+0.46%	+0.32
2019-05-27	AMD	Computex 2019 CEO Keynote: Dr. Lisa Su	11.1%	+3.10	+9.44%	+1.73
2019-06-03	Apple	WWDC 2019 Keynote	19.4%	+2.16	-0.18%	-0.10
2019-06-10	AMD	AMD Next Horizon Gaming (E3 2019)	7.7%	+1.61	-3.20%	-0.59
2019-06-11	Tesla	2019 Annual Meeting of Stockholders	5.5%	-0.05	-1.38%	-0.27
2019-08-07	Samsung Electronics	Samsung Galaxy Unpacked August 2019 (Brookly	8.2%	+1.57	-2.25%	-1.55
2019-08-07	AMD	2nd Gen AMD EPYC Launch (EPYC Horizon)	3.3%	-0.29	+13.67%	+2.55
2019-09-10	Apple	By innovation only	15.8%	+1.25	+3.26%	+1.81
2019-09-25	Meta Platforms	Oculus Connect 6 (OC6)	6.7%	+0.57	-1.25%	-0.52
2019-10-02	Microsoft	Microsoft Surface event	9.1%	+2.55	+0.63%	+0.50
2019-10-15	Alphabet	Made by Google '19 (Pixel 4 event)	2.9%	-0.65	+1.04%	+0.58
2019-11-21	Tesla	Cybertruck Unveil	3.2%	-0.44	-6.15%	-1.55
2019-12-03	Amazon	AWS re:Invent 2019 - keynote (Andy Jassy)	1.8%	-0.31	-1.06%	-0.70
2020-01-06	AMD	AMD Press Conference at CES 2020	3.8%	-0.06	-1.94%	-0.50
2020-01-07	Tesla	Made-in-China Model 3 Delivery Ceremony and	13.6%	+1.32	+8.61%	+2.05
2020-02-11	Samsung Electronics	Galaxy Unpacked February 2020 (San Francisco	4.6%	+0.18	+0.51%	+0.42
2021-06-10	Tesla	Model S Plaid Delivery Event	2.0%	-0.65	-0.31%	-0.05
2021-08-19	Tesla	AI Day 2021	1.2%	-0.79	+1.29%	+0.24
2021-10-07	Tesla	2021 Annual Meeting of Stockholders	2.2%	-0.60	+1.10%	+0.24
2021-11-30	Amazon	AWS re:Invent 2021 - keynote (Adam Selipsky)	3.0%	+0.36	-0.02%	-0.01
2022-03-22	Tesla	Giga Berlin Opening and First Model Y Delive	18.8%	+2.19	+8.36%	+1.96
2022-04-07	Tesla	Cyber Rodeo (Giga Texas Grand Opening)	15.1%	+1.56	-4.44%	-1.08
2022-05-11	Alphabet	Google I/O 2022 Keynote	8.9%	+1.33	+0.82%	+0.56
2022-08-04	Tesla	2022 Annual Meeting of Stockholders	8.7%	+0.49	-5.70%	-1.28

2022-09-30	Tesla	AI Day 2022	1.9%	-0.67	-17.49%	-3.86
2022-10-06	Alphabet	Made by Google '22 (Pixel 7 event)	0.4%	-1.49	+2.45%	+1.38
2022-11-10	AMD	together we advance_data centers (4th Gen EP	3.9%	-0.02	+6.57%	+2.01
2022-11-29	Amazon	AWS re:Invent 2022 - keynote (Adam Selipsky)	2.0%	-0.20	-1.94%	-0.75
2022-12-01	Tesla	Tesla Semi Delivery Event	4.5%	-0.22	-2.76%	-0.64
2023-01-04	AMD	CES 2023 Keynote: Dr. Lisa Su	6.6%	+1.15	-3.15%	-0.96
2023-02-01	Samsung Electronics	Galaxy Unpacked February 2023 (San Francisco	4.9%	+0.28	+2.89%	+2.11
2023-02-07	Microsoft	Microsoft AI event: new Bing and Edge	1.0%	-1.23	+3.66%	+2.47
2023-03-01	Tesla	2023 Investor Day	1.0%	-0.81	-5.85%	-1.28
2023-05-10	Alphabet	Google I/O 2023 Keynote	8.2%	+1.09	+8.10%	+3.79
2023-05-16	Tesla	2023 Annual Meeting of Stockholders	7.8%	+0.33	+2.63%	+0.55
2023-05-23	Microsoft	Build 2023 opening keynote	4.0%	+0.18	+0.02%	+0.01
2023-05-29	NVIDIA	NVIDIA COMPUTEX 2023 keynote	2.5%	-0.10	-1.86%	-0.58
2023-06-13	AMD	AMD Data Center and AI Technology Premiere	4.6%	+0.28	-2.74%	-0.87
2023-07-26	Samsung Electronics	Galaxy Unpacked July 2023 (Seoul)	4.9%	+0.26	+0.42%	+0.31
2023-09-20	Amazon	Amazon Devices & Services Event 2023	1.3%	-0.64	-2.20%	-0.83
2023-09-21	Microsoft	Microsoft event: Copilot and Surface	5.2%	+0.72	+1.27%	+0.69
2023-09-27	Meta Platforms	Meta Connect 2023	6.1%	+0.36	+0.13%	+0.03
2023-10-04	Alphabet	Made by Google '23 (Pixel 8 event)	2.9%	-0.66	+0.99%	+0.43
2023-11-15	Microsoft	Microsoft Ignite 2023 opening keynote	4.2%	+0.28	+1.23%	+0.72
2023-11-28	Amazon	AWS re:Invent 2023 - CEO keynote (Adam Selip	2.0%	-0.19	-1.17%	-0.51
2023-11-30	Tesla	Cybertruck Delivery Event	1.1%	-0.79	-4.44%	-0.98
2023-12-06	AMD	AMD Advancing AI (2023)	6.1%	+0.93	+7.51%	+2.18
2024-01-17	Samsung Electronics	Galaxy Unpacked January 2024 (San Jose)	5.7%	+0.58	+3.70%	+2.60
2024-03-18	NVIDIA	GTC 2024 keynote	4.6%	+1.40	-1.26%	-0.37
2024-05-14	Alphabet	Google I/O 2024 Keynote	5.4%	+0.18	-0.16%	-0.08
2024-05-20	Microsoft	Introducing Copilot+ PCs	4.4%	+0.34	+1.63%	+1.16
2024-05-21	Microsoft	Build 2024 opening keynote	3.9%	+0.13	+1.21%	+0.86
2024-06-02	NVIDIA	NVIDIA COMPUTEX 2024 keynote	2.1%	-0.37	+5.19%	+1.72
2024-06-03	AMD	Computex 2024 Opening Keynote: Dr. Lisa Su	4.9%	+0.39	-4.76%	-1.41
2024-06-13	Tesla	2024 Annual Meeting of Stockholders	5.8%	+0.00	+1.89%	+0.45
2024-07-10	Samsung Electronics	Galaxy Unpacked July 2024 (Paris)	6.0%	+0.70	-3.56%	-2.28
2024-08-13	Alphabet	Made by Google '24 (Pixel 9 event)	4.5%	-0.13	-3.99%	-1.94
2024-09-25	Meta Platforms	Meta Connect 2024	8.3%	+1.13	+0.24%	+0.08
2024-10-10	AMD	AMD Advancing AI 2024	2.6%	-0.59	-2.49%	-0.70
2024-10-10	Tesla	We, Robot	0.1%	-0.96	-10.57%	-2.46
2024-11-19	Microsoft	Microsoft Ignite 2024 keynote	2.4%	-0.60	-0.37%	-0.33
2024-12-03	Amazon	AWS re:Invent 2024 - CEO keynote (Matt Garma	2.3%	-0.03	+2.62%	+1.51
2025-01-06	AMD	AMD Press Conference at CES 2025	3.2%	-0.35	+3.28%	+0.92
2025-01-06	NVIDIA	NVIDIA CES 2025 keynote	3.9%	+0.90	-4.00%	-1.10
2025-01-22	Samsung Electronics	Galaxy Unpacked January 2025 (San Jose)	1.9%	-0.87	-0.45%	-0.23
2025-03-18	NVIDIA	GTC 2025 keynote	3.6%	+0.68	-1.63%	-0.43
2025-04-04	Microsoft	Microsoft 50th Anniversary + Copilot event	6.2%	+1.20	+3.16%	+2.32
2025-05-19	NVIDIA	NVIDIA COMPUTEX 2025 keynote	1.7%	-0.67	-0.25%	-0.06
2025-05-19	Microsoft	Build 2025 opening keynote	3.2%	-0.20	+1.23%	+0.87
2025-05-20	Alphabet	Google I/O 2025 Keynote	6.6%	+0.56	+3.53%	+1.77
2025-05-21	AMD	AMD Press Conference at Computex 2025	0.9%	-1.35	+1.07%	+0.34
2025-06-11	NVIDIA	GTC Paris 2025 keynote at VivaTech	3.5%	+0.63	+0.57%	+0.15
2025-06-12	AMD	AMD Advancing AI 2025	3.8%	-0.09	-2.21%	-0.70
2025-07-09	Samsung Electronics	Galaxy Unpacked July 2025 (Brooklyn)	3.0%	-0.48	+2.32%	+1.17
2025-08-20	Alphabet	Made by Google '25 (Pixel 10 event)	7.7%	+0.91	-0.21%	-0.09
2025-09-17	Meta Platforms	Meta Connect 2025	8.6%	+1.27	-1.25%	-0.55
2025-09-30	Amazon	Amazon Devices & Services Event 2025	8.1%	+3.40	-1.78%	-0.92
2025-10-28	NVIDIA	GTC Washington, D.C. 2025 keynote	2.3%	-0.28	+7.50%	+2.45
2025-11-06	Tesla	2025 Annual Meeting of Shareholders	5.7%	-0.01	-4.25%	-0.94
2025-11-11	AMD	AMD Financial Analyst Day 2025	0.6%	-1.48	+5.86%	+1.86
2025-11-18	Microsoft	Microsoft Ignite 2025 opening keynote	2.0%	-0.76	-3.74%	-2.49
2025-12-02	Amazon	AWS re:Invent 2025 - CEO keynote (Matt Garma	2.4%	+0.03	-1.30%	-0.70
2026-01-05	NVIDIA	NVIDIA Live at CES 2026 keynote	1.3%	-0.97	-0.12%	-0.04
2026-01-05	AMD	CES 2026 Opening Keynote: Dr. Lisa Su	3.1%	-0.37	-6.03%	-1.44
2026-02-25	Samsung Electronics	Galaxy Unpacked February 2026 (San Francisco	3.4%	-0.29	+2.96%	+1.40
2026-03-16	NVIDIA	GTC 2026 keynote	2.6%	-0.01	-1.38%	-0.60

2026-05-19	Alphabet	Google I/O 2026 Keynote	7.7%	+0.91	-2.97%	-1.41
2026-05-20	Tesla	Model S & X Signature Delivery Event	1.8%	-0.67	+1.16%	+0.32
2026-06-01	NVIDIA	NVIDIA GTC Taipei 2026 keynote (COMPUTEX 202	3.0%	+0.27	+4.78%	+2.16
2026-06-02	Microsoft	Build 2026 opening keynote	3.7%	+0.02	-6.57%	-3.53
2026-07-22	Samsung Electronics	Galaxy Unpacked July 2026 (London)	5.2%	+0.42	-2.72%	-1.20
2026-07-23	AMD	AMD Advancing AI 2026	4.0%	-0.01	-2.97%	-0.57
2026-08-12	Alphabet	Made by Google '26 (Pixel 11 event)	3.8%	-0.36	-0.34%	-0.15

Appendix B. Reproducing the results

The repository runs end to end with `make all`: catalogue validation, audio download (cached, never committed), detection, caption download and detector validation, applause features, market data, the event study, the pre-registered analysis, figures, notebooks and this report. Derived per-second timelines are committed, so every number in this report can be recomputed from `make features study analyze figures report` without downloading any audio.